

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

PLANT CHECKLIST

Organics Plant Controls Walk-Around Checklist

A one-page checklist for composting and anaerobic digestion (AD) sites. Print it, walk the plant, and mark what you find. It pairs with the full guide on keeping control systems reliable in composting and AD plants. Nothing here asks you to defeat a safety function; if an interlock is nuisance-tripping, note it and find the cause, do not bypass it.

Gas safety comes before the clipboard. H₂S kills fast and deadens your sense of smell around 100 ppm (olfactory fatigue; IDLH roughly 100 ppm), so your nose quits right as it turns lethal. Methane is explosive in air from about 5 percent (LEL) to 15 percent (UEL). Digesters, gas holders, sumps, wet wells, and pump vaults are permit-required confined spaces — test the atmosphere before entry, wear a bump-tested personal 4-gas monitor (H₂S, LEL, O₂, CO), know your wind and escape direction, and never enter a gas structure solo.

Before touching anything by hand, prove it is at a zero-energy state. Lock out and tag out (LOTO) and verify de-energization before you check a blower damper or linkage, a mixer, or a pump by hand — a remote start or a spring-loaded actuator does not care that you are standing in it.

Enclosures and the corrosive environment

- ☐ Control panels sealed, gasketed, and doors latched tight
- ☐ Positive pressure / clean-air purge on panels in process air, and it is actually flowing
- ☐ Unused gland holes and cable entries sealed
- ☐ Panel heater keeping interior above the dew point; no condensation inside
- ☐ Enclosure cooler / heat exchanger working on the hot side
- ☐ No visible tarnish (black/gray) on terminals, relays, or connector pins
- ☐ Corrosion-inhibitor sachets present and not spent

Sensors and field devices

- ☐ Pile temperature probes reading plausibly; checked against a handheld reference
- ☐ Oxygen and moisture sensors calibrated and logged for drift
- ☐ Gas analyzers (digester, biofilter) within calibration

- ☐ Thermowells clear of debris and packing
- ☐ Any sensor reading a suspiciously steady or flat value flagged for check
- ☐ Drift log kept so a failing sensor is caught before it lies

Aeration and blower control (ASP / in-vessel)

- ☐ Blower loop settled, not hunting up and down
- ☐ Control strategy matched to batch stage (oxygen early, heat later); active-pile oxygen roughly in the 5-15% band (confirm against your own process)
- ☐ VFDs cool, clean, and not nuisance-tripping
- ☐ Dampers and linkages free, not worn or seized (LOTO and zero-energy verified before checking by hand)
- ☐ Reversing / valve sequencing (if used) confirmed correct
- ☐ Biofilter / scrubber differential pressure (dP) trended; not climbing toward clogged
- ☐ Biofilter media moisture in range

Anaerobic digester process

- ☐ Feed metered steadily, not slugged; loading rate ramped gradually
- ☐ Leading indicator trended (volatile fatty acids to alkalinity ratio, if available)
- ☐ Mixing homogeneous without over-shearing; no crust or grit build-up
- ☐ Mixers interlocked with level and foam detection
- ☐ Digester temperature steady in its range (mesophilic or thermophilic)
- ☐ Heat exchanger approach temperature trended; watching for fouling

Safety interlocks (verify present and healthy, never defeat)

- ☐ Digester over- and under-pressure protection functional
- ☐ Pressure / vacuum relief devices clear and in service
- ☐ Air-ingress guarded: gas-blower low-pressure cutout and vacuum relief sized/set so the blower cannot pull the gas space into a vacuum and draw air in
- ☐ Biogas area electrical equipment and wiring match the area classification (Class I Div 1/2 or Zone 0/1/2, NEC 500/505); any NFPA 496 pressurization alarms/de-energizes on loss of pressure
- ☐ Flame arrestors and flare / gas-use interlocks working
- ☐ Gas detection (methane, H₂S) calibrated and alarming correctly
- ☐ Confined-space atmosphere tested before any entry to digester, gas holder, sump, wet well, or pump vault
- ☐ Pump low-level / dry-run protection active

- ☐ High-temperature limits on heating loop active
- ☐ Any nuisance trip logged with its root cause, not bypassed

Chasing intermittent faults

- ☐ Event log / high-speed capture set up to timestamp faults and surrounding I/O
- ☐ Fault times plotted against ambient temperature and humidity
- ☐ Faults clustering at dawn (coldest, dampest) noted
- ☐ Suspect terminations reflowed; shields and grounds checked
- ☐ Signal and power cabling separated; shields landed at one end
- ☐ VFD / contactor noise ruled in or out with evidence

SCADA, odor, compliance, and records

- ☐ Long-term trends kept: gas flow/composition, feed, pile temp/oxygen, blower energy, biofilter dP, heating loop
- ☐ Pathogen-reduction (PFRP, 40 CFR Part 503) time-at-temperature captured with timestamps; ~55 C / 131 F held for the duration your specific process requires (in-vessel/ASP vs windrow differ — confirm the exact time and turnings)
- ☐ Odor control and flare/emissions uptime logged for the permit
- ☐ Historian survives power blips and network hiccups
- ☐ Compliance report is a query, not a paper scramble
- ☐ Alarm list rationalized; every alarm has a meaning and an action

When the answer is not software

- ☐ Clogged media, worn dampers, cavitating pumps addressed mechanically
- ☐ Corroded input cards / sensors replaced, not coded around
- ☐ Obsolete, unsupportable controller flagged for a planned migration

Field notes from Strule Automation, independent and vendor-neutral controls engineering, Bay Area and Northern California. Want a second set of eyes on any line above? Get in touch: struleautomation.com or info@struleautomation.com